Student Name(s)/Group: Group 3 **Score:** 70/75

Metric	Max. Possible Score	Score	Comments
Quality of written Report	10 pts	08	-A few formatting or graph set up errors (e.g income vs smoking bar graph).
Design and Methodology of the Study	20 pts	20	
Statistical Analyses and Interpretation of Software Outputs	35 pts	33	- See annotations in document
Conclusion and Implications	10 pts	10	Ok
Individual contribution and presentation	25 pts		
Total Score	100 pts		

NOTE: Individual contribution is based on the self/peer assessment, ~~presentation~~, and my observation of your role in the project (e.g., communication with me out of class and/or contribution during the 4 group meetings held in class).

Smoking

Not informative enough. See "title/topic" in project description.

[redacted]

Abstract

This study uses survey data on smoking habits from the UK. The data set can be used for analyzing the demographic characteristics of smokers and types of tobacco consumed. The data set includes a data frame with 1691 observations on the following 12 variables. The variables include age, marital status, highest qualification, nationality, ethnicity, income, region, amt_weekends, weekdays, etc. we summarized the distribution with a box plot, and a density plot. These findings can illuminate if there is a difference in the average number of cigarettes smoked per day between males and females.

Introduction

People's health habits are shaped by many factors, some being their background, lifestyle, and financial situation. One behavior researchers often study is smoking, and how it may differ between males and females and their different income levels. Some research suggests that smoking patterns can vary depending on a person's sex and how much income they earn. Looking at the relationship between sex, income, and smoking can help us better understand why certain groups may be more likely to smoke. Understanding these patterns can help public health experts create better strategies to reduce smoking and improve overall health.

Research Questions

1. Is there a difference in the average number of cigarettes smoked per day between males and females?
2. Do people with less income smoke more cigarettes?

Literature Review

The article "Are there sex differences in tobacco smoking?" was written on Jan 1st, 2020. It highlights that generally, men tend to use all tobacco products at higher rates than women. In 2015, 16.7 percent of adult males and 13.6 percent of adult females smoked cigarettes. Such differences may relate to a combination of physiological (particularly ovarian hormones), cultural, and behavioral factors. Research suggests that women experience stronger cravings than men in response to stress, but men may be more responsive to environmental cues.

The article "Gender and age differences among current smokers in a general population survey," written by U John and others, was published on June 3rd, 2005. The author claims that the proportions of current smokers tended to be larger among female than among male ever smokers aged 40 or above. Women, compared to men, showed adjusted odds ratios of 1.7 to 6.9 at ages 40 to 90 or older in contrast to men. No such interaction existed for age of onset of smoking or cigarettes per day.

The article "Association between marital status and cigarette smoking: Variation by Race and Ethnicity," written by Michael W. Ramsey Jr. and others and written on February 2nd, 2020, highlights and explains that the 2015 National Health Interview Survey showed that the prevalence of current cigarette smoking was 13.1%, 16.6%, and 20.0% among adults who are married/living with partners, single, and divorced/separated/widowed, respectively. Married adults benefit from receiving spousal support and coping resources that are unique to the marriage union. Proving that smoking prevalence is consistently higher among divorced, separated, or never-married individuals compared to married adults. In the U.S. now, roughly 21-45% of separated/divorced adults smoke, compared to 13-27% of married adults. Married adults are more likely to quit or never smoke due to social support.

The article "US Cigarette Smoking Disparities by Race and Ethnicity—Keep Going and Going!" by René A. Arrazola and others was written on June 1, 2023. Talks about the results from 2011 to 2020: linear decreases in prevalence and population estimates were observed for non-Hispanic White (20.6% to 13.3%; 32.1 million to 20.7 million), non-Hispanic Black (19.4% to 14.4%; 5.1 million to 4.0 million), and Hispanic (12.9% to 8.0%; 4.2 million to 3.3 million) adults. For American Indians, prevalence remained around 27%, and a linear increase in the population estimate was observed from 400,000 to 510,000. Relative disparity did not change across racial and ethnic categories.

The article "Adult Cigarette Smoking Rates by State" tells us how smoking prevalence in the U.S. varies by region, with the highest rates in the Midwest (14.0%) and South (12.4%) and lower rates in the Northeast (10.4%) and West (8.9%). Key states like West Virginia (23.8%) and Kentucky (23.6%) have the highest prevalence, while Utah (7.9%) and California (10.0%) have the lowest.

Methodology

Sampling procedures and Data Collection

The dataset used in this study comes from the OpenIntro "Smoking" dataset, which examines whether individuals smoke and, if so, how much they smoke during weekdays and weekends. The dataset includes participants ranging in age from 16 to 97 years old and records several demographic characteristics such as gender, marital status, education level, ethnicity, gross income, region, and type of cigarette consumed.

The dataset contains 1,691 observations and 12 variables. For the purposes of this study, the analysis will primarily focus on the relationship between gender and income and how these factors may relate to smoking behavior.

The data were originally collected through a survey administered to over 1,500 participants. Because the data come from survey responses, participants voluntarily provided their answers about their smoking habits and characteristics. While the exact sampling procedure is not stated, it is likely that a non-random or convenience sampling method was used.

Data Analysis and Results

Research Question 1: Is there a difference in the average number of cigarettes smoked per day between males and females?

To answer the first research question, the relationship between gender and average cigarettes smoked per day was examined using both numerical summaries and visualizations. The variable gender is a categorical variable, while cigarette consumption was measured using the numerical variables amt_weekdays and amt_weekends, which represent the number of cigarettes smoked on weekdays and weekends. A new variable called average cigarettes smoked per day was created using a weighted average formula to combine weekday and weekend smoking levels.

First, a summary statistics table was created showing the sample size, mean, median, and standard deviation of cigarettes smoked per day for males and females. The results showed that males smoked an average of 16.6 cigarettes per day, while females smoked an average of 12.9 cigarettes per day. The median number of cigarettes smoked per day was 15 for males and 12.6 for females, indicating that males tend to smoke more cigarettes per day on average.

Next, a boxplot was used to visualize the distribution of cigarettes smoked per day by gender. The boxplot shows that the median smoking level for males is higher than for females and that the distribution for males includes several higher outliers. A density plot was also created to compare the distribution of cigarette consumption between the two groups. The density curves show that the distribution for males is generally shifted toward higher cigarette consumption values compared to females.

Overall, both the numerical summaries and the visualizations suggest that there is a relationship between gender and cigarette consumption. In this dataset, males tend to smoke more cigarettes per day on average than females.

Research Question 2: Do people with less income smoke more cigarettes?

In order to answer the second research question, the relationship between gross income and average cigarettes smoked per day was analyzed. The variable gross income is a categorical variable representing different income groups, while the variable average cigarettes smoked per day is a numerical variable representing cigarette consumption.

First, summary statistics were calculated for each income category, including the sample size, mean, median, and standard deviation of cigarettes smoked per day. These statistics allow comparison of smoking behavior across different income groups.

Next, a bar chart was created showing the average number of cigarettes smoked per day for each income category. The bar chart provides a visual comparison of smoking levels across income groups.

The results show that cigarette consumption varies across income levels, but the pattern is not completely consistent. Some lower-income groups appear to have relatively high average cigarette consumption, while other groups have lower averages. Because the pattern is not strictly decreasing as income increases, the results do not show a strong or consistent relationship between income level and cigarette consumption in this dataset.

Overall, the analysis suggests that while cigarette consumption differs between income groups, there is not clear evidence that people with lower income consistently smoke more cigarettes than people with higher income in this dataset.

Research question 1

Is there a difference in the average number of cigarettes smoked per day between males and females?

```
library(dplyr)
library(ggplot2)

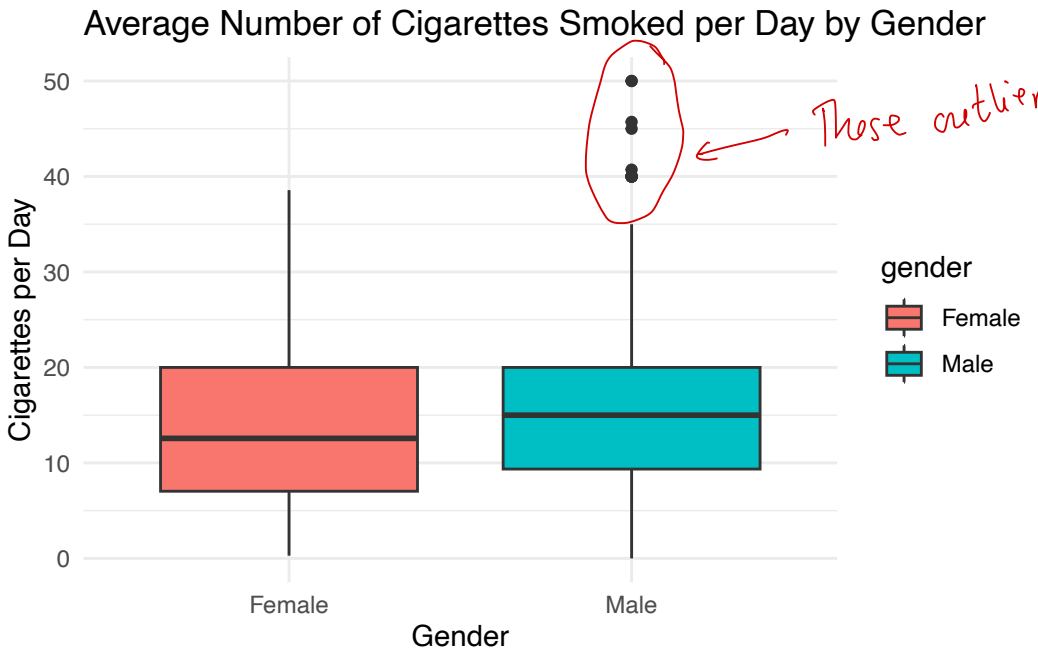
smoking2 <- smoking %>%
  mutate(avg_cig_per_day = (5 * amt_weekdays + 2 * amt_weekends) / 7)

ggplot(smoking2, aes(x = gender, y = avg_cig_per_day, fill = gender)) +
  geom_boxplot() +
  labs(
    title = "Average Number of Cigarettes Smoked per Day by Gender",
    x = "Gender",
```

Hide the code

```
    y = "Cigarettes per Day"
  ) +
  theme_minimal()
```

```
Warning: Removed 1270 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



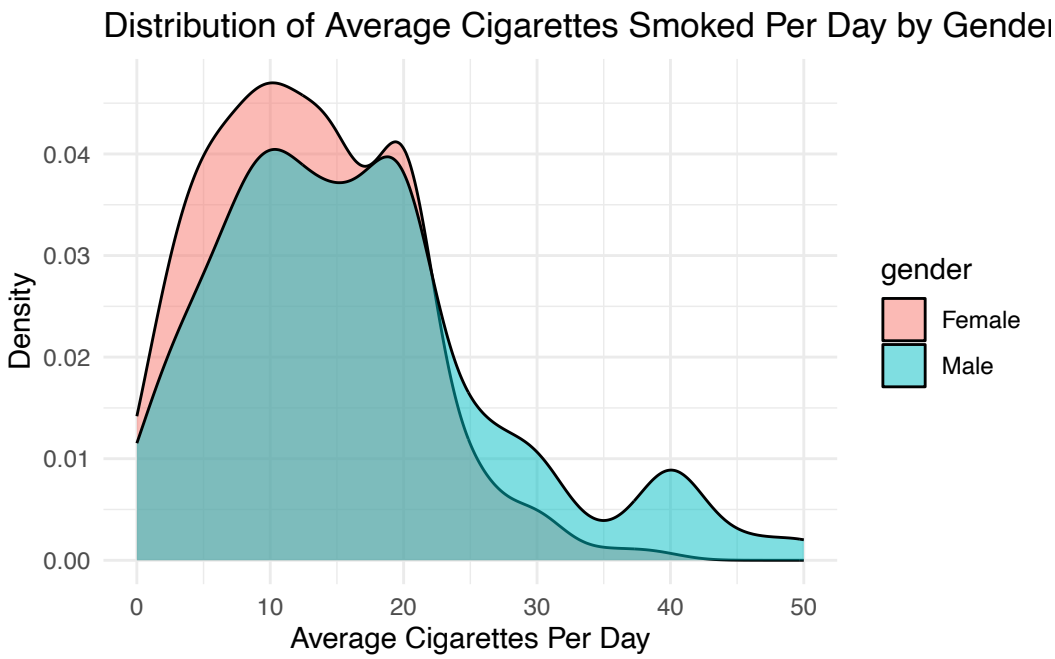

```
library(dplyr)
library(ggplot2)

smoking2 <- smoking %>%
  mutate(avg_per_day = (5 * amt_weekdays + 2 * amt_weekends) / 7)

ggplot(smoking2, aes(x = avg_per_day, fill = gender)) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Distribution of Average Cigarettes Smoked Per Day by Gender",
    x = "Average Cigarettes Per Day",
    y = "Density"
  ) +
  theme_minimal()
```

Hide code.

```
Warning: Removed 1270 rows containing non-finite outside the scale range
(`stat_density()`).
```




```r
library(dplyr)

smoking_table <- smoking %>%
  mutate(avg_per_day = (5 * amt_weekdays + 2 * amt_weekends) / 7) %>%
  group_by(gender) %>%
  summarise(
    Sample_Size = n(),
    Mean_Cigarettes_Per_Day = mean(avg_per_day, na.rm = TRUE),
    Median_Cigarettes_Per_Day = median(avg_per_day, na.rm = TRUE),
    Std_Dev = sd(avg_per_day, na.rm = TRUE)
  )

smoking_table
```

```
# A tibble: 2 x 5
  gender Sample_Size Mean_Cigarettes_Per_Day Median_Cigarettes_Per_Day Std_Dev
  <fct>        <int>                   <dbl>                     <dbl>   <dbl>
1 Female         965                    12.9                      12.6    7.34
2 Male           726                    16.6                      15     10.6
```


Software output analyses not sufficiently discussed.

Research question 2

Do people with less income smoke more cigarettes?

```
library(dplyr)
library(ggplot2)

smoking2 <- smoking %>%
  mutate(avg_cig_per_day = (5 * amt_weekdays + 2 * amt_weekends) / 7)
summary_income <- smoking2 %>%
  group_by(gross_income) %>%
  summarise(
    sample_size = n(),
    mean_cigs = mean(avg_cig_per_day, na.rm = TRUE),
    median_cigs = median(avg_cig_per_day, na.rm = TRUE),
    sd_cigs = sd(avg_cig_per_day, na.rm = TRUE)
  )

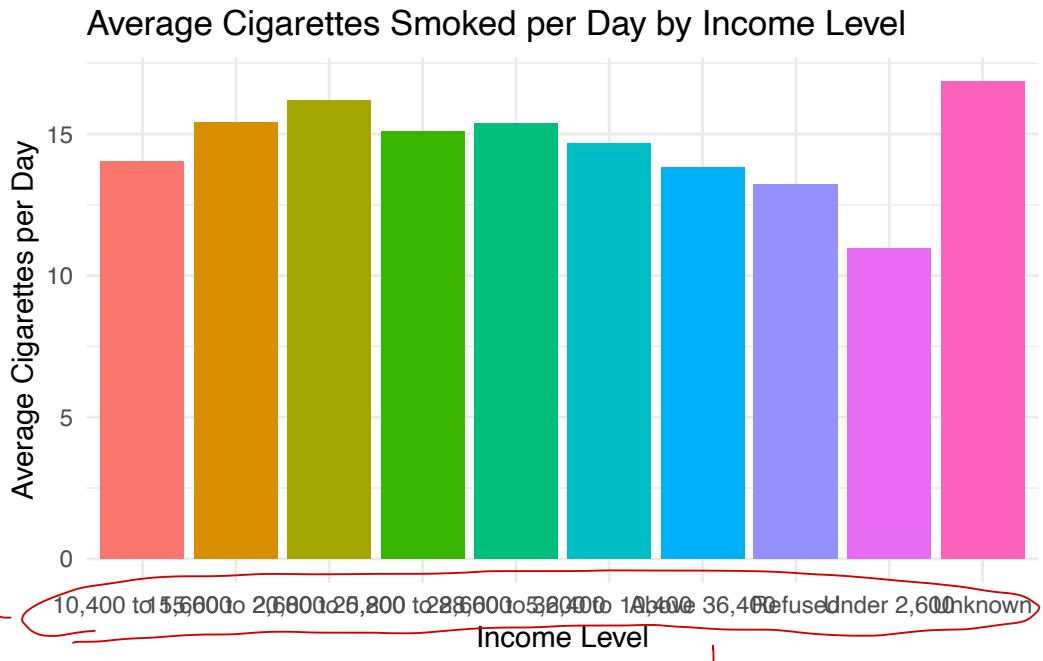
summary_income
```

```
# A tibble: 10 x 5
   gross_income     sample_size mean_cigs median_cigs sd_cigs
   <fct>                  <int>     <dbl>       <dbl>   <dbl>
 1 10,400 to 15,600         268      14.0        12.9    7.89
 2 15,600 to 20,800         188      15.4        15      9.08
 3 2,600 to 5,200           257      16.2        15     11.3
 4 20,800 to 28,600         155      15.1        15      8.35
 5 28,600 to 36,400          79      15.4        15     12.6
 6 5,200 to 10,400          396      14.7        14.3    9.29
 7 Above 36,400              89      13.8        12     11.8
 8 Refused                  108      13.2        10      7.65
 9 Under 2,600              133      11.0        10.7    6.12
10 Unknown                   18      16.9        15      7.39
```

```
mean_income <- smoking2 %>%
  group_by(gross_income) %>%
  summarise(mean_cigs = mean(avg_cig_per_day, na.rm = TRUE))

ggplot(mean_income, aes(x = gross_income, y = mean_cigs, fill = gross_income)) +
  geom_col() +
  labs(
    title = "Average Cigarettes Smoked per Day by Income Level",
```

```
    x = "Income Level",
    y = "Average Cigarettes per Day"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



It would help if you place income level on the y-axis and average cig on x-axis.

Conclusion

The results of this study show that smoking behavior can vary based on both gender and income level. From the analysis, males tended to smoke slightly more cigarettes per day on average compared to females. This finding generally supports previous research discussed in the literature review, which suggested that men often use tobacco products at higher rates than women.

The analysis of income levels showed some variation in smoking behavior across different income groups. Some of the lower income categories had higher average cigarette use, while higher income groups generally showed slightly lower averages. However, the relationship was not perfectly consistent across every category, suggesting that income alone may not fully explain smoking behavior. Other factors such as stress, lifestyle, or access to health education may also influence how much people smoke.

Overall, the findings of this study mostly align with the trends discussed in previous research. Smoking behavior appears to be influenced by a combination of demographic and social factors rather than just one variable. While this study provides some insight into how gender and income relate to smoking habits, there are limitations to consider. Because the data came from survey responses and likely used a non-random sample, the results may not represent the entire population. Future research could explore additional variables such as age, education, or marital status to better understand the patterns behind smoking behavior.

References

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